

Additions Manifest for a Tagalog-English Catastrophic Forgetting Paper

Paul Pajo
Independent Researcher
paulamerigo.pajojr@benilde.edu.ph

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Abstract

This is not the P1.1 results paper and not a table of invented English-continuation scores. It is the complete specification that turns Pajo (2026) — equal-exposure nanochat depth versus held-out Tagalog bits-per-byte on WikiText-TL-39, AsPredicted #306780 — into a *new* catastrophic-forgetting (CF) study relative to English, with a new paper, a new GitHub repository, and a new Hugging Face repository. Twenty-seven additions are stratified into six layers: explicit, implicit, inferred, extrapolated, residual, hidden. Each addition names the P1.1 anchor (including Table 2 BPB values), the CF-paper section it lands in, the GitHub artifact, the Hub artifact, whether it is blocking, and the pass/fail test. A super-granular procedure then orders registration, corpus, pilots, confirmatory runs, close-out, paper writing, and publication so that a second operator can execute without improvising. Critical path: extra-seed matrix $\rightarrow D_{1x}$ English pilots $\rightarrow$ new AsPredicted PDF $\rightarrow$ English-native from-scratch baseline $\rightarrow$ full phase-2 grid. Do not amend #306780. Do not upload phase-2 weights into pageman/nanochat-filipino-p1-fixed-d20-3x. Do not treat Hub metadata val_bpb (262,144-token loop) as the before-score.

How to read this document

Frozen parent anchors (do not recompute)

Master table of 27 additions

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- E1. Multiple seeds (blocking)
- E2. Split label on every CF caption (blocking)
- E3. BOS-bestfit crop asymmetry (blocking)
- E4. Native-speaker ratings remain absent (non-blocking, must declare)
- E5. No downstream classification / CORE / SFT as primary (blocking)

Layer 2 — Implicit additions (I1–I5)

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- I2. Two-phase training protocol and learning-rate lock (blocking)
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- N3. 0.01 BPB threshold inheritance
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Unified experimental grid (after PDF)

Super-granular procedure

- Gate P2-0 — Freeze the parent (read-only)
- Gate P2-1 — Create empty P2 GitHub repository
- Gate P2-2 — Create empty P2 Hugging Face repository
- Gate P2-3 — English and news corpus audit (I1, H1, H2)
- Gate P2-4 — D_{1x} English pilots (R5)
- Gate P2-5 — File the new AsPredicted (H5)
- Gate P2-6 — English-native from-scratch EN0 (H6)
- Gate P2-7 — Confirmatory phase-2 grid
- Gate P2-8 — Seal C , then optional test (H3, I3)
- Gate P2-9 — Secondary methods (X1–X3), optional
- Gate P2-10 — Paper compilation
- Gate P2-11 — Publish GitHub and Hub close-out

Paper section ↔ addition map

GitHub README claim / not-claim (copy text)

What remains out of paper 1

Per-addition operator card (copy into the ledger)

JSON schemas (GitHub manifests/ and results/)

- seed_matrix.json (E1, R1)
- source_manifest_en.json (I1)
- budget_phase2.json (I2)
- delta_tagalog.json (I3)
- test_access_log_p2.json (H3)
- category.json (X5)

Command templates (do not improvise flags)

Hugging Face README body (paste)

Paper paragraph inserts (do not paraphrase into hype)

Gate file checklist (tick in the ledger, not in chat)

Compute sketch (not a budget promise)

Failure modes mapped to addition IDs

Status

Parent paper: https://www.researchgate.net/publication/412302216_Equal-Exposure_Depth_and_Held-Out_Tagalog_Bits-per-Byte_on_WikiText-TL-39
Parent code: <https://github.com/pageman/nanochat-filipino>
Parent weights: <https://huggingface.co/pageman/nanochat-filipino-p1-fixed-d20-3x>
Stage-1 CF manuscript: docs/papers/p2-cf-english/paper.tex
This manifest: docs/papers/p2-cf-english/additions-manifest.tex
Intended new code repo: pageman/nanochat-filipino-p2-cf-en
Intended new Hub repo: pageman/nanochat-filipino-p2-cf-en

How to read this document

A naive CF checklist (“add EWC, add replay, measure accuracy drop”) would be a wrong answer here. P1.1 is not an English classifier. It is a Tagalog language-modeling measurement with four locked depths, a common $D_{\text{actual}} = 19,267,584$ tokens, a reconstructed split label, a 0.01 BPB interpretation rule, and a one-test-touch clock. Every addition below is anchored to that architecture.

Wrong answers, named so they can be rejected on sight:

1. Describe generic CF methodology without Table 2’s exact BPB values (d8 1.179135, d12 1.180824, d16 1.195546, d20 1.172248).
2. Ignore equal exposure as the structural asset, or call the parents “trained to convergence” (they were not; R_d is 0.458 at d8 and 0.044 at d20).
3. Treat “English” as a monolith: no file, no raw-versus-Moses decision, no token-count basis, no domain control.
4. Skip the registration-authority chain (new PDF > this manifest > P2 ledger > deviation card).
5. Reuse P1.1’s one `test_bpb=1.164768` as a post-English number.
6. Write new weights into the parent Hub repository.
7. Use loop `val_bpb` from `meta_000294.json` (d20 1.117213) as $\text{BPB}_{\text{before}}$.
8. Retrain the tokenizer after seeing English fertility.
9. Conflate forgetting, collapse, and retention without predeclared bins.
10. Put Design B (Cebuano) or SFT into the first confirmatory table.

Layer confidence, as used while writing: explicit 99%; residual 95%; implicit 90%; inferred 85%; extrapolated 80%; hidden 75%. Hidden items are the most load-bearing and the easiest to miss.

Frozen parent anchors (do not recompute)

Quantity	d8	d12	d16	d20
val_bpb_full (DV)	1.179135	1.180824	1.195546	1.172248
train_bpb_full	0.836702	0.528818	0.458045	0.672393
val—train gap	0.342432	0.652006	0.737501	0.499855
untrained val	3.289109	3.289358	3.289354	3.289106
byte unigram val	4.453225	4.453225	4.453225	4.453225
Hub loop val_bpb	1.124545	1.125139	1.137399	1.117213
P_{scaling}	41.9M	110.1M	234.9M	435.2M
$R_d = D_{3x}/P_{\text{scaling}}$	0.457834	0.174413	0.081756	0.044129
checkpoint SHA prefix	9c407f4f	5dfccc27	525301eb	9e30fff3

Shared: nanochat 92d63d4e8bb4df75c3b71618f31ddde2378b2bcd; tokenizer 04436b85...bc5a8; token_bytes.pt a5dbc1c8...2132; $T = 2048$; $B = 65,536$; $N = 294$; split reconstructed_article_70_15_15; packed val target bytes 5,868,797; one parent test read on d20 only. Extra-seed sample SD at d8 ≈ 0.010318 (val only; not used to pick D^*).

Master table of 27 additions

ID	Layer	One-line name	Blocking?	First artifact
E1	Explicit	Multi-seed phase-2 matrix	Yes	manifests/seed_mat
E2	Explicit	Split label on every CF caption	Yes	caption template
E3	Explicit	BOS-bestfit crop accounting	Yes	crop audit JSON
E4	Explicit	No native-speaker ratings	No	ethics paragraph
E5	Explicit	No downstream classifiers	Yes	exclusions list
I1	Implicit	English corpus/split/tokenizer	Yes	source_manifest_en
I2	Implicit	Two-phase train + LR lock	Yes	phase2_train.yaml
I3	Implicit	$\Delta\text{BPB}_{\text{tag}} + 0.01$ rule	Yes	analysis plan
I4	Implicit	Forward-transfer English BPB	Yes	results/en_val.json
I5	Implicit	Joint-training ceiling (A3)	Yes	arm A3 run cards
N1	Inferred	Depth-stability hypothesis	Yes	AsPredicted box
N2	Inferred	Equal-exposure as CF control	Yes	intro paragraph
N3	Inferred	Inherit 0.01 as Δ threshold	Yes	glossary
N4	Inferred	R_d as plasticity index	Yes	Table R_d vs Δ
X1	Extrapolated	EWC baseline (secondary family)	After A2	arm-EWC/
X2	Extrapolated	Experience-replay baseline	After A2	arm-ER/

ID	Layer	One-line name	Blocking?	First artifact
X3	Extrapolated	Stability–plasticity scatter	After A2	figure spec
X4	Extrapolated	BWT/FWT sign conventions	Yes	metric card
X5	Extrapolated	Collapse/forgetting/retention bins	Yes	category card
R1	Residual	Extra-seed matrix reuse	Yes	copy P1.1 seeds
R2	Residual	Post-phase-2 document bootstrap	No	bootstrap script
R3	Residual	Mid-run Tagalog BPB trajectory	Diagnostic	eval-every log
R4	Residual	gzip entropy normalization	No	gzip JSON
R5	Residual	D_{1x} English pilots	Yes	pilot run cards
H1	Hidden	Tagalog-news domain control	Yes	arm A4 spec
H2	Hidden	Tokenizer fertility asymmetry	Yes	fertility table
H3	Hidden	Second Tagalog test-read clock	Yes	test_access_log_p2
H4	Hidden	Null P1.1 as CF asset	Yes	intro reframing
H5	Hidden	New AsPredicted, not an amendment	Yes	new PDF
H6	Hidden	English-native from-scratch baseline	Yes	arm EN0

Critical path (do not reorder): R1/E1 seed matrix → R5 English D_{1x} pilots → H5 CF registration PDF → H6 English-native baseline → I1–I5 + H1 full phase-2 grid → X1–X3 secondary methods → paper/GitHub/Hub close-out.

X1–X3 are *not* the first confirmatory table. They start after untreated A2–A1 exists, unless the new PDF names them confirmatory (not recommended for paper 1).

Layer 1 — Explicit additions (E1–E5)

These are things P1.1 already names as missing or constrained. A CF paper that ignores them repeats the parent limitation as if it were unknown.

E1. Multiple seeds (blocking)

P1.1’s four-depth table is one seed (torch 42). Extra seeds 1–2 exist only at d8/d12 on validation, sample $SD \approx 0.010$ at d8, which already swallows the 0.006887 d20–d8 parent gap. CF deltas will sit in that noise. Specification: continuation seeds $\{0, 1, 2\}$ on arms A1 and A2 at all four depths ($4 \times 2 \times 3 = 24$ confirmatory trainings). A0 and A3 at seed 0 unless budget allows parity. GitHub: manifests/seed_matrix.json with parent

seed versus continuation seed. Hub: `d{8,12,16,20}/a2/seed{0,1,2}/`. Pass: three finite Tagalog `val_bpb_full` values per cell; sample SD reported; no ranking if the seed interval of $A_2 - A_1$ contains 0.

E2. Split label on every CF caption (blocking)

P1.1 never recovered 2019 files. Label `reconstructed_article_70_15_15` must appear on every Tagalog retention table, including after English training. The English split gets its own label, e.g. `wikitext103raw_article_70_15_15`, never mixed into Tagalog captions. GitHub: caption lint in `scripts/p2/check_captions.py`. Pass: zero Tagalog tables without the reconstructed label.

E3. BOS-bestfit crop asymmetry (blocking)

Packed Tagalog val scores 5,868,797 target bytes versus 6,771,275 raw UTF-8 (13.3% below). English documents have a different length distribution, so crop rate will change. If packing is unchanged, Tagalog packed-byte count after phase 2 must still equal 5,868,797 or the evaluator moved. GitHub: `results/packing_audit.json` with `n_cropped_fills` for Tagalog and English. Pass: Tagalog packed bytes identical to parent; English crop rate reported, not “fixed” after seeing Δ .

E4. Native-speaker ratings remain absent (non-blocking, must declare)

P1.1 invented no 1–5 ratings. CF samples after English continuation will look like mixed Taglish or broken English. Do not score them. Ethics paragraph only. Hub: no sample leaderboard.

E5. No downstream classification / CORE / SFT as primary (blocking)

P1.1 excluded CORE, chat, dengue, hate-speech, NLI. CF paper primary remains BPB. Permuted-MNIST-style accuracy is not this study. GitHub exclusions YAML. Pass: no CORE flag other than -1; no SFT data in confirmatory trains.

Layer 2 — Implicit additions (I1–I5)

Structurally required by any two-phase LM CF design, never spelled as a CF protocol in P1.1.

I1. English corpus, split, and tokenizer specification (blocking)

English is not “some English.” Lock: WikiText-103 *raw* (not Moses @-@ WT103) so both languages are encyclopedia prose. Canonical: LF only. Split: SHA-256 of UTF-8, 70/15/15, independent of Tagalog. Tokenizer: *parent Tagalog 32,768 BPE only*. Token budget counted as $\sum \text{len}(\tau(d))$ under that BPE, not GPT-2 tokens, not words. Record file SHA, $T_{\text{en,train}}$, fertility (bytes/token) on English train/val. If raw WT103 is unavailable, stop and file a deviation *before* using tokenized WT103. Do not substitute TLUnified, OSCAR, ClimbMix, or C4 as the confirmatory English stream (domain shift stacked on

language). GitHub: manifests/source_manifest_en.json, split_manifest_en.json, fertility_en.json. Hub README: English source hash, not the text.

I2. Two-phase training protocol and learning-rate lock (blocking)

Phase 1 is finished: load model_000294.pt only. Parent optimizer states were never published, so Adam moments cannot resume. Phase 2: fresh optimizer; $T = 2048$; $B = 65,536$; $N = 294$; $D_{\text{phase2}} = 19,267,584$; --num-iterations 294 explicit; --target-param-data-ratio = parent R_d for that depth, never -1 ; CORE off; device-batch 8, halve only for VRAM. Peak LR = $0.3 \times$ parent scheduled peak (from meta_000294.json or trainer formula). Warmup $\min(40, \lfloor 0.05N \rfloor) = 14$. This is continuation, not a second pretrain. GitHub: configs/phase2.yaml, command generator mirroring scripts/p1/generate_gate_i_runcards.py. Hub: each run folder stores the exact command string.

I3. Forgetting metric $\Delta\text{BPB}_{\text{tag}}$ and threshold (blocking)

$$\Delta\text{BPB}_{\text{tag}}(d, a, s) = \text{val_bpb_full}_{\text{after}}(d, a, s) - \text{val_bpb_full}_{\text{P1.1}}(d).$$

Before-score is Table 2, not Hub loop val. Confirmatory contrast:

$$C(d, s) = \text{val_bpb_full}(d, A2, s) - \text{val_bpb_full}(d, A1, s).$$

Positive C means English hurt Tagalog more than extra Tagalog did. Material if $|\bar{C}| \geq 0.01$. Evaluator: copy scripts/p1/gate_j_full_bpb.py byte-identical except output paths. GitHub: results/delta_tagalog.json. Pass: parent before-scores match Table 2 to printed precision.

I4. Forward-transfer / plasticity metric (blocking)

Retention without acquisition is incomplete. Secondary: English val_bpb_full after A2 (and A3, EN0). Not used to pick depth. Same evaluate_bpb, English val JSONL, $T = 2048$. GitHub: results/en_val.json. Sign convention: lower English BPB is better plasticity; do not subtract it from Tagalog Δ into one scalar.

I5. Joint-training upper bound (blocking)

Arm A3: 50/50 *document* mix of Tagalog train and English train, same D_{phase2} , shuffled at document level, never concatenating a Tagalog article to an English article inside one packed row as a silent curriculum. A3 is a ceiling on retention under mixed exposure, not a proof of “no forgetting.” GitHub: mixer script with seed. Pass: token counts 50/50 $\pm$ documented remainder $< B$.

Layer 3 — Inferred additions (N1–N4)

Follow-on logic from Table 2’s actual pattern, not from a generic “deeper is more stable” slogan.

N1. Depth-stability hypothesis (preregister as directional, not as a ranking)

Table 2 train–val gaps: d8 0.342, d12 0.652, d16 0.738, d20 0.500. The widest gap is d16, not d20; d20 is the `val_bpb_full` minimum by 0.006887 over d8. Inferred directional guess for *untreated* English forgetting:

$$\bar{C}(d16) > \bar{C}(d20) > \bar{C}(d8)$$

i.e. mid-depth most unstable, deepest not automatically most stable, shallowest least overfit and maybe most robust. This is a hypothesis to put in the new AsPredicted box. It is *not* a license to rank depths if gaps < 0.01 or seed intervals overlap. Alternative reading: R_d not gap predicts $\bar{C}$; then $\bar{C}$ should fall as R_d rises (d8 most data-rich, least plastic). Pre-register *which* of gap-order vs R_d -order is primary; the other is secondary. Recommendation: primary = R_d -order test (d8 vs d20 contrast of $\bar{C}$); gap-order is specified secondary because it uses P1.1’s most distinctive shape.

N2. Equal-exposure as CF control (intro load-bearing paragraph)

Most CF depth claims confound size with extra pretraining tokens. P1.1 removed that confound for phase 1. The CF intro must say: parents saw the same 19.27M tokens; therefore a later depth effect on $\bar{C}$ cannot be dismissed as “d20 simply trained longer in phase 1.” Residual: they are still not equally optimized (R_d differs). Caption that residual every time N2 is invoked.

N3. 0.01 BPB threshold inheritance

P1.1 used 0.01 so a 0.006887 dip was not a ranking. CF inherits 0.01 as the material-forgetting threshold for $\bar{C}$ and for d20–d8 of $\bar{C}$, unless the new PDF names another number *before* phase-2 BPB. Do not tighten to 0.001 after seeing a small $\bar{C}$.

N4. R_d as plasticity index

Plot $\bar{C}$ against $R_d \in \{0.458, 0.174, 0.082, 0.044\}$. If $|\bar{C}|$ rises as R_d falls, the story is residual undertraining, not “depth is a stability dial.” GitHub: `results/rd_vs_C.json`. Hub model card: one sentence that R_d is not a hyperparameter sweep.

Layer 4 — Extrapolated additions (X1–X5)

What the CF canon expects if this paper is to be readable next to EWC/GEM/ER. These are integration points, not replacements for A2–A1.

X1. EWC baseline (secondary family, after untreated A2)

Kirkpatrick et al. elastic weight consolidation on the parent, then English A2-budget . Fisher estimated on Tagalog train only, λ predeclared (recommend a two-value grid $\{10, 100\}$ as exploratory unless PDF says otherwise). Same tokenizer, T , B , N . Do not tune λ on Tagalog val after seeing C . GitHub: `scripts/p2/ewc_train.py`, `arm-EWC/`. Hub: separate folders, `parent_sha256` pinned. Not confirmatory for paper 1 unless the PDF says so.

X2. Experience replay (secondary)

Replay buffer = frozen Tagalog train documents. Mix ratios $\{0, 0.05, 0.20, 0.50\}$ of phase-2 tokens, remainder English, total still D_{phase2} . Ratio 0 is A2; do not double-count. Selection of ratio is not allowed from Tagalog test. This is Design D; paper 1 may report seed-0 d20 only as exploratory.

X3. Stability–plasticity scatter

x = Tagalog $\Delta\text{BPB}_{\text{tag}}$ (A2; higher x = more forgetting). y = English val BPB (higher y = worse acquisition). One point per depth $\times$ seed. Joint A3 and EN0 as reference symbols. No Pareto claim without the 0.01 rule.

X4. BWT and FWT with BPB sign conventions

Standard accuracy BWT is after-minus-before on old tasks (negative = forgetting). For BPB, *positive* after-minus-before on Tagalog is forgetting. Define:

$$\begin{aligned}\text{BWT}_{\text{BPB}} &= \text{val_bpb_full}^{\text{tag}}(\text{after}) - \text{val_bpb_full}^{\text{tag}}(\text{P1.1}), \\ \text{FWT}_{\text{BPB}} &= \text{val_bpb_full}^{\text{en}}(\text{A2}) - \text{val_bpb_full}^{\text{en}}(\text{EN0}).\end{aligned}$$

FWT uses the English-native from-scratch model (H6) as the reference, not a random init. Negative FWT_{BPB} means the Tagalog parent helped English (forward transfer). Publish the formulas in a metric card so reviewers do not flip signs.

X5. Category pre-registration: retention / forgetting / collapse

Bins on Tagalog A2 val_bpb_full relative to parent:

- **Retention:** $\bar{C} < 0.01$ and no collapse flag.
- **Forgetting:** $\bar{C} \geq 0.01$ and not collapse.
- **Collapse:** after A2, Tagalog val BPB $\geq$ parent untrained, or ≥ 4.453225 , or $\Delta \geq 1.0$.

A run cannot be titled “collapse” because $\Delta = 0.02$. GitHub: `results/category.json`.

Layer 5 — Residual additions (R1–R5)

P1.1 suppressed or sidelined items that become CF assets.

R1. Extra-seed matrix

P1.1 extra seeds at d8/d12 are secondary for D^* and primary for noise-scale planning. Copy the protocol: validation only, no test, same evaluator. Extend to d16/d20 *before* claiming CF seed needs. If P1.1 extra-seed files exist only on a stopped pod, treat them as unavailable and rerun rather than inventing SD.

R2. Document-level bootstrap after phase 2

P1.1 bootstrap used per-document non-overlap, not BOS-bestfit; all four 95% CIs overlapped (~ 1.23 – 1.39). Re-run the same bootstrap on post-A2 Tagalog val as a secondary packing, labeled `not_val_bpb_full`. Do not pick depth from it.

R3. Mid-run Tagalog BPB trajectory during English training

P1.1 forbade selecting D^* from step-200 d12 min 1.084991. CF still *records* Tagalog val every 50 steps at `eval-tokens=262144` during A2 as a diagnostic curve of when forgetting happens. Primary remains final-step full BPB. GitHub: `results/trajectories/` with an explicit `not_primary: true` flag.

R4. gzip entropy normalization

P1.1 gzip `—9` on raw val UTF-8 was 2.739 bits/byte. After A2, gzip of Tagalog val does not change (data are frozen). gzip is not a forgetting metric. Optional: gzip of English val as a compressor baseline for plasticity, never as BWT.

R5. D_{1x} English pilots (blocking on the critical path)

P1.1 D_{1x} pilots (step 98) showed d8/d12 worse than D_{3x} . Before spending 24 confirmatory A1/A2 runs, run d8 and d20, seed 0, $N = 98$, English only, Tagalog `val_bpb_full`. Purpose: wiring, VRAM, fertility shock, not C . If Tagalog BPB becomes NaN, stop. GitHub: `docs/run-cards/p2-pilots/`. These pilots must finish *or be waived by a dated deviation* before the confirmatory PDF claims the grid is feasible. They must not leak into the confirmatory table.

Layer 6 — Hidden additions (H1–H6)

Frame shifts P1.1 could suppress because it never left Tagalog Wikipedia. In a CF paper they become load-bearing.

H1. Language-versus-domain confound (blocking for the phrase “relative to English”)

A2 on English Wikipedia changes language *and* dump statistics. To isolate language, add arm A4: matched-budget continuation on **Tagalog news** (TLUnified or a predeclared news dump), same tokenizer, same D_{phase2} . Contrast of interest for “English caused it”:

$$C_{\text{lang}} = C_{A2} - C_{A4}.$$

If A4 forgets as much as A2, the cause is domain shift or extra training, not English. A4 is confirmatory for the title phrase “relative to English”; if compute forces a cut, the title must drop that phrase and say “relative to extra Tagalog Wikipedia” (A2–A1 only). Do not harvest TLUnified after seeing A2.

H2. Tokenizer fertility asymmetry (blocking)

Parent BPE was trained on Tagalog only. P1.1 val fertility ≈ 4.53 bytes/token versus GPT-2 ≈ 2.76 . On English, the same BPE will over-segment (“shred”) English into shorter tokens, raising English token counts for the same UTF-8 bytes and making English acquisition look harder than a GPT-2-tokenized English LM. That is part of the intervention. Report English bytes/token before any phase-2 claim. Forbidden fix: train a bilingual tokenizer after seeing English BPB. Optional later paper: matched-fertility byte-level models.

H3. Test-set authority chain and a second Tagalog test read (blocking)

P1.1: `test_read_events=1`, d20 only, 1.164768, SHA 3bd19345.... A CF paper that wants a Tagalog *test* Δ needs a **new** predeclared event: one test read after all validation C are sealed, on a predeclared subset (recommend d20 seed-0 A1 and A2 only, or all depths seed-0 A2 only — pick in the PDF). Clock: `test_access_log_p2.json` starts at 0. P1.1’s 1.164768 is the parent before-test for d20 only; other depths have no parent test BPB and cannot have a test Δ without a new parent test, which P1.1 forbade. Therefore confirmatory Tagalog test Δ is d20-only, or the PDF must say “validation only, no test Δ .” Recommended: validation C is primary; one d20 A2 test BPB is secondary; no test for d8/d12/d16.

H4. Null-result reframing (intro)

P1.1’s non-ranking at 0.01 is the CF paper’s strongest asset: depth was not a performance dial, so a later depth effect on C is new information rather than “bigger models were already better.” Write that paragraph before methods. Do not reopen D^* .

H5. Separate AsPredicted / ResearchBox (blocking)

New PDF, new box. Authority: new PDF > this manifest > docs/papers/p2-cf-english/paper.tex > P2 ledger > deviation. Filing occurs after R5 pilots show feasibility and before any confirmatory A2 `val_bpb_full`. Draft: docs/papers/p2-cf-english/aspredicted-draft-p2.txt (extend it with H1/H6/N1 before filing).

H6. English-native from-scratch baseline (blocking for “relative to English”)

The title phrase is not only “Tagalog model then English data.” It also requires an English-native model trained *from scratch* on the same English stream, same D_{phase2} , same Tagalog BPE (fair tokenizer), same depths, seed 0: arm EN0. Plasticity FWT compares A2 English BPB to EN0. A Tagalog parent that reaches EN0 English BPB has transferred; one that is far worse has paid a tokenizer-and-initialization tax. EN0 is not a parent of forgetting (it never saw Tagalog). GitHub: `runs/en0-d{8,12,16,20}/`. Hub: `en0/` not inside `p1-fixed-d20-3x`.

Unified experimental grid (after PDF)

Confirmatory family (paper 1):

Arm	Parent	Phase-2 stream	Role
A0	P1.1 d	none	Eval drift
A1	P1.1 d	Tagalog Wiki train	Continued-training drift
A2	P1.1 d	English WT103-raw train	Language+dump intervention
A3	P1.1 d	50/50 documents	Joint ceiling
A4	P1.1 d	Tagalog news train	Domain control (H1)
EN0	random	English WT103-raw train	English-native baseline (H6)

Primary contrast: $\bar{C}(d) = \overline{A2 - A1}$ over seeds $\{0, 1, 2\}$. Language isolation: $\overline{A2 - A4}$ at seed 0 if A4 is confirmatory. Plasticity: A2 English BPB versus EN0. Secondary after freeze: EWC, ER ratios.

Super-granular procedure

Execute in order. A later gate may not start if the previous gate’s JSON status is not pass. Deviation cards cannot make a departed run confirmatory after seeing BPB.

Gate P2-0 — Freeze the parent (read-only)

1. Verify local or Hub SHA-256 for `d8/d12/d16/d20 model_000294.pt` against §2.
2. Verify `tokenizer.pkl` and `token_bytes.pt` hashes.
3. Copy `scripts/p1/gate_j_full_bpb.py` into the new repo without editing packing math.
4. Write `PARENT.md` with ResearchGate URL, GitHub URL, Hub URL, commit `92d63d4`, AsPredicted `#306780`, and “do not amend.”

5. Do not run Tagalog test BPB. Do not touch test.jsonl.

Gate P2-1 — Create empty P2 GitHub repository

1. `gh repo create pageman/nanochat-filipino-p2-cf-en --public`.
2. Orphan first commit (same discipline as P1.1 public-main): no passcodes, no transfer/, no test.jsonl, no .pt weights.
3. Root files: LICENSE (MIT), LICENSE-RESEARCH.md, DATA_AND_CODE_NOTICES.md (WikiText-TL-39 + WikiText-103 + news dump if A4), PARENT.md, README.md (claim / not-claim / Quick Start / bibtex / DeepWiki / parent links).
4. Tree:

```
p2-cf-en/  
  README.md PARENT.md LICENSE LICENSE-RESEARCH.md  
  DATA_AND_CODE_NOTICES.md  
  configs/phase2.yaml  
  docs/PROTOCOL-p2-cf-en.md  
  docs/papers/ (this manifest + Stage-1 paper)  
  docs/run-cards/p2-pilots/  
  manifests/ (seed, source_en, split_en, fertility, budget_phase2,  
              gate_ledger, test_access_log_p2,  
  selection_record_p2)  
  patches/ (copy NANOCHAT_DATA_DIR.patch)  
  reproducibility/  
  results/ (empty JSON schemas with nulls)  
  scripts/p2/
```
5. .gitignore: *.pt, *.pkl, *.parquet, artifacts/, vendor/nanochat/, passcode files.
6. Push orphan main only. Do not force-push parent history that contains secrets.

Gate P2-2 — Create empty P2 Hugging Face repository

1. `hf repo create pageman/nanochat-filipino-p2-cf-en --type model --private` then make public after card review, or public immediately if card has no secrets.
2. YAML: license: other; languages tl, en; tags nanochat, catastrophic-forgetting, bits-per-byte; datasets linkanjarad/Wikitext-TL39 plus the locked English id.
3. README **Paper** / **Code** / **Parent** block:

```
parent_base_repo: pageman/nanochat-filipino-p1-fixed-d20-3x  
parent_paper:  
  https://www.researchgate.net/publication/412302216_...  
parent_code: https://github.com/pageman/nanochat-filipino  
this_code: https://github.com/pageman/nanochat-filipino-p2-cf-en
```
4. Folder schema (create .gitkeep only until weights exist):

d8/a0/ d8/a1/seed0/ d8/a2/seed{0,1,2}/ d8/a3/ d8/a4/ d8/en0/
... same for d12 d16 d20
ewc/ er/ # secondary, later

5. Each future weight folder: model_*.pt, meta_*.json, SHA256, command string. No optimizer states required. No Tagalog test.jsonl.
6. Do not push into p1-fixed-d20-3x.

Gate P2-3 — English and news corpus audit (I1, H1, H2)

1. Download WikiText-103 raw; hash; LF-only canonical; article reconstruct; 70/15/15 hash split; write split_manifest_en.json.
2. Encode train with frozen parent BPE; write $T_{\text{en,train}}$, bytes/token, compare to Tagalog 4.53.
3. If A4 is confirmatory: lock Tagalog news source, same pipeline, split_manifest_news.json. News val/test never enter Tagalog Wikipedia val/test.
4. Place shards via NANOCHAT_DATA_DIR hook. Never python -m nanochat.dataset.
5. Write-protect English test JSONL; keep out of active train dir.

Gate P2-4 — D_{1x} English pilots (R5)

1. Host card: named NVIDIA GPU, image, torch, device-batch.
2. d8 seed0 A2 $N = 98$; d20 seed0 A2 $N = 98$.
3. Tagalog val_bpb_full once at end (full evaluator), plus loop logs.
4. Pass: finite BPB, hashes unchanged, no test read, VRAM fit at $T = 2048$.
5. Fail: stop confirmatory; deviation or smaller device-batch only (not smaller T).

Gate P2-5 — File the new AsPredicted (H5)

1. Expand aspredicted-draft-p2.txt with N1 (which depth order is primary), H1 (A4 yes/no), H6 (EN0 required), H3 (test policy), X5 bins, 0.01 rule, equal D_{phase2} , frozen tokenizer.
2. File PDF. Store SHA-256. ResearchBox deposit without passcode in git.
3. Set ledger registration.url. No confirmatory A2 full BPB before this timestamp.

Gate P2-6 — English-native from-scratch EN0 (H6)

1. From random init, depths 8 and 20 seed 0 at minimum; 12 and 16 if budget.
2. Same English train stream, $N = 294$, Tagalog BPE, $T = 2048$.
3. English val BPB only until sealed; English test at most once later.
4. Upload to Hub en0/d8, en0/d20. GitHub manifests only.

Gate P2-7 — Confirmatory phase-2 grid

1. Serial or documented-parallel: for $d \in \{8, 12, 16, 20\}$, A0 (eval only), A1 seeds 0–2, A2 seeds 0–2, A3 seed 0, A4 seed 0 if confirmatory.
2. After each run: finite check; SHA of weights; Tagalog full val BPB; do not look at Tagalog test.
3. Write per-cell JSON: nats, bytes, excluded specials, packed bytes (must match parent on Tagalog).
4. A0 must recover Table 2 within documented host noise; if not, stop (evaluator/host drift).

Gate P2-8 — Seal C , then optional test (H3, I3)

1. Compute $\bar{C}(d)$ and sample SD from A1/A2 seeds. Write `selection_record_p2.json` with `test_read_count=0`.
2. Apply 0.01 rule and X5 bins. Do not drop depths.
3. If PDF allows: one Tagalog test read on predeclared cells; append `test_access_log_p2.json`; never use test to choose depth.
4. English test at most once, after English val sealed, not mixed with Tagalog test in one command.

Gate P2-9 — Secondary methods (X1–X3), optional

Only after Gate P2-8 JSON exists. EWC/ER cannot rewrite C . Label exploratory unless the PDF already included them (it should not, for paper 1).

Gate P2-10 — Paper compilation

1. Fill Stage-1 docs/papers/p2-cf-english/paper.tex Table phase-2 with real numbers; do not edit P1.1 Table 2.

2. Intro uses H4+N2. Methods use I1–I5, H2, H6. Results: $C(d)$, A4 if run, EN0, scatter X3, categories X5.
3. Limitations: single parent seed, R_d residual, fertility shredding, reconstructed split, packing crop.
4. Acknowledgements may match P1.1. Cheng is not coauthor.
5. Build tex, md, txt, html, docx, pdf. Cite parent ResearchGate URL.
6. Do not claim “deeper is always more stable.” Do not claim P1.1 showed forgetting.

Gate P2-11 — Publish GitHub and Hub close-out

1. GitHub: results JSON, run cards, deviation cards, paper outputs (no test.jsonl).
2. Hub: upload only new run folders; model card copies Δ table and points at GitHub results/ for audit; parent SHAs listed; loop vs full BPB distinction restated.
3. Tag Hub revision. Record commit in final_checkpoint_manifest_p2.json.
4. DeepWiki link after GitHub is public:
<https://deepwiki.com/pageman/nanochat-filipino-p2-cf-en>.

Paper section $\leftrightarrow$ addition map

CF paper section	Additions that must appear
Title / abstract	H4, H5, H6, I3, X5, N3
Introduction	H4, N2, N1 (as hypothesis), parent Table 2 cite
Related work	X1, X2, X4, McCloskey/French/Luo
Registration	H5, H3, E1, N3
English data	I1, H1, H2
Phase-2 train	I2, E3, R5, R3 (diagnostic)
Metrics	I3, I4, X4, X5, N4
Arms	I5, H1, H6, A0–A4, EN0
Results	E1, N1 test, X3, R2 optional
Limitations	E2, E4, E5, R_d , fertility, reconstructed split
Ethics / lineage	parent URLs, no passcode, no parent-Hub overwrite

GitHub README claim / not-claim (copy text)

Claim: This repository is the close-out archive for a preregistered equal-exposure continuation study: frozen P1.1 Tagalog nanochat parents, matched D_{phase2} , Tagalog val_bpb_full contrast A2–A1, optional language-versus-news control, English-native from-scratch reference.

Not-claim: Not P1.1. Not a chat model. Not CORE. Not proof that deeper models forget less. Not an amendment of AsPredicted #306780. Not a dump of test.jsonl. Checkpoints live on Hugging Face pageman/nanochat-filipino-p2-cf-en. Parent weights stay at pageman/nanochat-filipino-p1-fixed-d20-3x.

What remains out of paper 1

Design B (Cebuano/Ilocano/Hiligaynon). PackNet / GEM as confirmatory. SFT arm E. Changing tokenizer. ClimbMix. Reopening D^* . Using loop val as BWT. Native-speaker scores. Downstream NER/GCE as forgetting metrics.

Per-addition operator card (copy into the ledger)

For each ID, the operator writes one JSON object before the next ID starts. Fields: id, layer, started_at_utc, ended_at_utc, operator, status $\in$ {not_started, pass, fail, deferred}, artifacts[], hashes{}, notes, confirmatory (bool). Deferred is allowed only for X1, X2, X3, R2, R4, and for A4 if the PDF dropped the phrase “relative to English.” Deferred is not allowed for H5, I3, E1, I1, H2, H3, H6 if the title keeps “relative to English.”

JSON schemas (GitHub manifests/ and results/)

seed_matrix.json (E1, R1)

Keys: parent_seed (always 0 / torch 42), continuation_seeds [0, 1, 2], arms_requiring_three_seeds [A1, A2], arms_seed0_only [A0, A3, A4, EN0] unless upgraded, n_confirmatory_trainings (24 if A1/A2 $\times$ 4 depths $\times$ 3 seeds).

source_manifest_en.json (I1)

Keys: dataset_id, revision, raw_not_moses (must be true), file_sha256, n_rows, canonical LF_only, tokenizer_sha256 (must equal 04436b85...), t_en_train, bytes_per_token_train, bytes_per_token_val, gpt2_bytes_per_token_val (descriptive only).

budget_phase2.json (I2)

Keys: N 294, B 65536, D_phase2 19267584, token_unit parent_tagalog_bpe, lr_peak_fraction 0.3, warmup 14, fresh_optimizer true, never_ratio_minus_one true, core_metric_every -1.

delta_tagalog.json (I3)

Per depth and seed: bpb_before (copied from Table 2, not recomputed as a new truth), bpb_a0, bpb_a1, bpb_a2, bpb_a3, $C = a2 - a1$, $\Delta a2 = a2 - \text{before}$, packed_bytes (must be 5868797 on Tagalog val), n_excluded, finite.

test_access_log_p2.json (H3)

Starts test_read_events: 0. Each event: at_utc, kind (tagalog_test_bpb or english_test_bpb), depth, arm, seed, bpb, file_sha256, used_to_select: false. Forbidden: reading P1.1 test without a new event; using P1.1's 1.164768 as after.

category.json (X5)

Per depth: C_bar, C_sd, $\text{bin} \in \{\text{retention}, \text{forgetting}, \text{collapse}\}$, collapse_flags (untrained, unigram, delta_ge_1).

Command templates (do not improvise flags)

Parent load is always the Hub or local SHA-pinned model_000294.pt. Phase-2 train (A2 shown; A1 swaps data dir to Tagalog train shards; A4 to news shards):

```
python -m scripts.base_train \
  --device-type=cuda --depth=DEPTH --max-seq-len=2048 \
  --device-batch-size=8 --total-batch-size=65536 \
  --num-iterations=294 \
  --target-param-data-ratio=PARENT_RD \
  --eval-tokens=262144 --eval-every=50 --core-metric-every=-1 \
  --sample-every=200 --save-every=50 --warmup-steps=14 \
  --run=p2-a2-dDEPTH-sSEED --model-tag=p2-a2-dDEPTH-sSEED
```

Load-from-checkpoint flag must be whatever the pinned nanochat commit actually implements; record the exact flag in the run card *before* launch. If the trainer cannot load a custom .pt, stop and write a deviation; do not reimplement load_model ad hoc after seeing loss.

Tagalog full eval after train (validation phase only):

```
python scripts/p2/gate_j_full_bpb.py --phase val \
  --depth DEPTH --model-tag TAG --checkpoint-step FINAL
```

--phase test is illegal until test_access_log_p2.json says count 0 and selection_record_p2.json exists.

EN0 omits parent load, same English shards, same N, B, T , Tagalog BPE still mounted.

Hugging Face README body (paste)

Paste after YAML. Keep the loop-versus-full warning:

```
# nanochat-filipino-p2-cf-en
```

Paper (Stage 1 until numbers exist): see GitHub docs/papers/p2-cf-english/

Parent paper: https://www.researchgate.net/publication/412302216_...

Parent weights: pageman/nanochat-filipino-p1-fixed-d20-3x

Parent code: <https://github.com/pageman/nanochat-filipino>

This code: <https://github.com/pageman/nanochat-filipino-p2-cf-en>

This repo is continuation and English-native runs, not the P1.1 parents.

Do not rank forgetting from meta_*.json val_bpb (eval_tokens=262144).

Primary: Tagalog val_bpb_full contrast A2-A1. Threshold 0.01 BPB.

Tokenizer is the frozen Tagalog 32,768 BPE. English is WikiText-103 raw.

No test.jsonl. No SFT. No ClimbMix. License field other.

Paper paragraph inserts (do not paraphrase into hype)

Intro (H4+N2). “Pajo (2026) showed that under D_{3x} on WikiText-TL-39, nanochat depths 8–20 are not a held-out Tagalog BPB ranking at 0.01 BPB (exact minimum at depth 20, margin 0.006887 to depth 8). That null is useful here: if depth later changes the English-continuation contrast C , the change is not a restatement of a phase-1 performance ranking. Phase-1 exposure was matched ($D_{\text{actual}} = 19,267,584$); residual R_d still differs and is reported with C .”

Methods tokenizer (H2). “English is encoded with the parent Tagalog BPE. Fertility on English is expected to be worse than GPT-2. Bytes/token are reported. The tokenizer is not retrained.”

Methods English-native (H6). “Arm EN0 trains the same depths from scratch on the English stream with the same BPE and D_{phase2} . FWT_{BPB} is A2 English val BPB minus EN0 English val BPB. EN0 never sees Tagalog and is not a forgetting parent.”

Results rule (N3+X5). “We report $\bar{C}$ and sample SD. Values below 0.01 are not a ranking. Collapse is reserved for the predeclared binary.”

Gate file checklist (tick in the ledger, not in chat)

P2-0: four weight SHAs, two tokenizer SHAs, PARENT.md, evaluator copy, zero test reads.
P2-1: public GitHub, orphan commit, gitignore, MIT + research notices, empty results schemas.
P2-2: Hub repo, YAML other, parent block, empty arm folders, no parent-repo push.
P2-3: English raw hash, split label, fertility table, write-protected English test, no ClimbMix.
P2-4: d8 and d20 $N = 98$ pilots, finite Tagalog val, host card, VRAM at $T = 2048$.
P2-5: new AsPredicted PDF SHA, ResearchBox without passcode in git, ledger URL.
P2-6: EN0 d8/d20 at least, English val sealed.
P2-7: 24 A1/A2 runs + A0/A3/(A4), packed bytes match, A0 recovers Table 2.
P2-8: C sealed at test count 0; optional one d20 test; categories assigned.
P2-9: EWC/ER only if labeled secondary.
P2-10: paper table filled, P1.1 table untouched, six formats built.
P2-11: GitHub results public, Hub weights in p2 repo only, DeepWiki link.

Compute sketch (not a budget promise)

P1.1 seed-0 D_{3x} wall times on A40 were about 0.027/0.061/0.119/0.204 hours for d8/d12/d16/d20. Phase-2 is the same N and B , plus eval. Order-of-magnitude: 24 A1/A2 trains $\approx 6 \times$ a four-depth seed-0 sweep if depths are balanced, plus EN0, A3, A4, pilots. Record actual hours on the host card. Do not cut d20 after seeing C .

Failure modes mapped to addition IDs

NaN Tagalog BPB after A2 $\rightarrow$ stop, E1/R5. Packed byte count changed $\rightarrow$ evaluator moved, E3. A0 $\neq$ Table 2 $\rightarrow$ host/eval drift, P2-7. English Moses @-@ used silently $\rightarrow$ I1 fail. Tokenizer retrained $\rightarrow$ H2 fail. Test read before seal $\rightarrow$ H3 fail. Weights pushed to parent Hub $\rightarrow$ lineage fail. C computed from loop val 1.117213 $\rightarrow$ I3 fail. Title says “relative to English” but A4 and EN0 absent $\rightarrow$ H1/H6 fail. Collapse claimed at $\Delta = 0.02 \rightarrow$ X5 fail. #306780 edited $\rightarrow$ H5 fail.

Status

This manifest is a specification. Phase-2 BPB does not exist here. Filling $C(d)$ before H5 is a protocol break.

9 P. Pajo. Equal-exposure depth and held-out Tagalog bits-per-byte on WikiText-TL-39. 2026. https://www.researchgate.net/publication/412302216_Equal-Exposure_Depth_and_Held-Out_Tagalog_Bits-per-Byte_on_WikiText-TL-39. S. Merity

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